

● $\mathcal{H}$: public structural library (background)

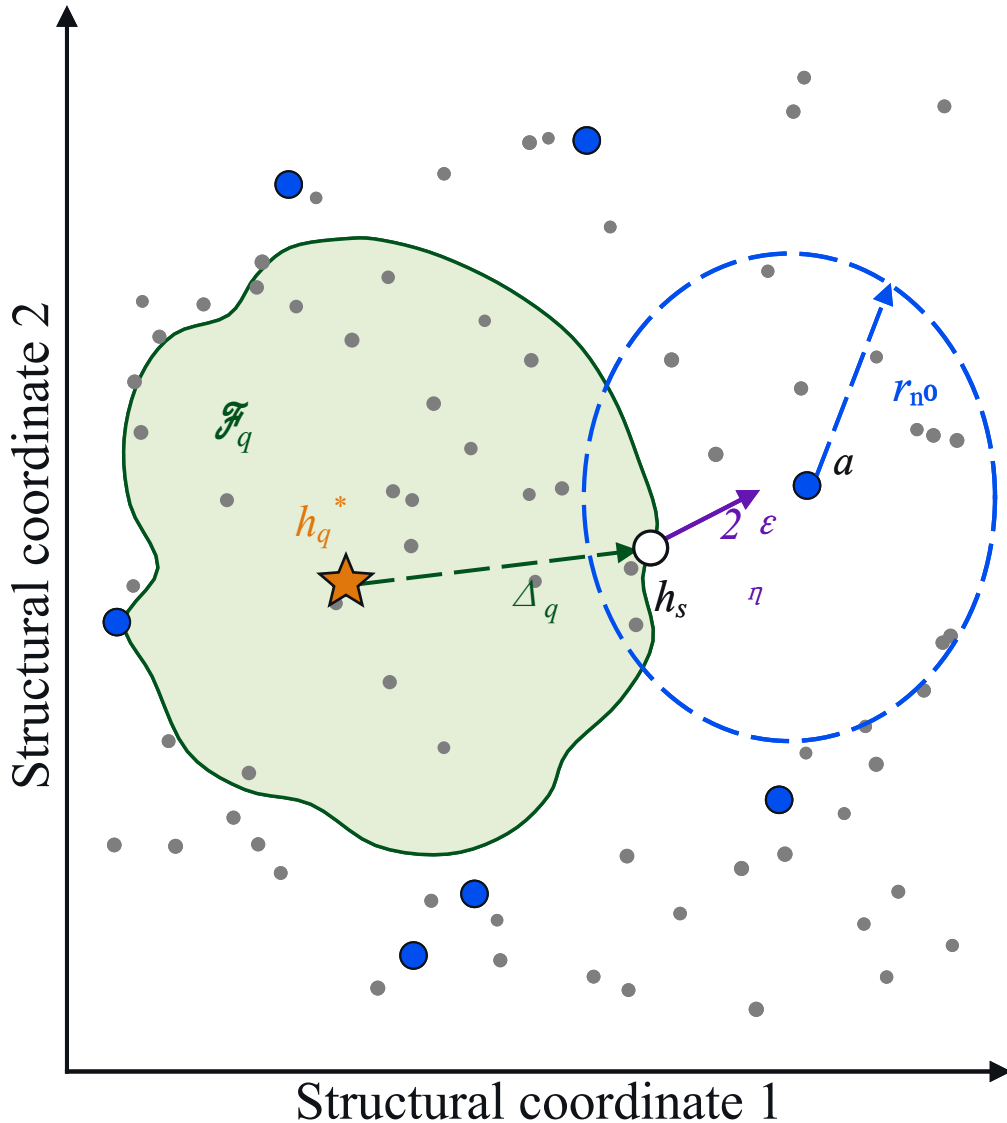
● $\mathcal{A}$: atlas members (selected)

○ $\mathcal{F}_q$: true feasible basin

★ h_q^* : safe center (target)

○ h_s : ideal aligned profile (related-mode center)

○ $B(a, r_{n0})$: coverage ball of atlas member a with radius r_{n0}



Coverage theorem

If

$$L_q (r_{n0} + \Delta_q + 2 \epsilon_\eta) \leq \gamma_q$$

then

$$\mathcal{A} \cap \mathcal{F}_q \neq \emptyset$$

The atlas intersects the true feasible basin; at least one atlas member lies in $\mathcal{F}_q$.

Interpretation of terms

Δ_q = task-aligned profile discrepancy (distance from h_q^* to h_s)

$2 \epsilon_\eta$ = coordinate mismatch error (ideal vs. learned/estimated center)

r_{n0} = atlas coverage radius (in structural coordinates)

γ_q / L_q = feasible depth in profile space (safety margin scaled by Lipschitz constant)



Takeaway: If the feasible basin is deep enough relative to task discrepancy, coordinate error, and atlas coverage radius, then at least one atlas member lies in the true feasible region, so fresh verification can succeed.